Impulse and Step Responses in a Buck Converter with a Current PI Controller

This paper is part of the Engineering and Technology Learning Portfolio, documenting continuous self-study and experimental work.

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Introduction

Impulse and step responses are fundamental tools for understanding how a dynamic system behaves when subjected to sudden changes in input. These responses describe transient characteristics, stability behavior, and final steady-state performance. In power electronics and control engineering, these principles directly apply to converters, filters, and closed-loop control systems.
A buck converter operating under a current PI controller is a practical example where classical first-order and second-order system theory translates directly into real hardware behavior.

Time Response Overview

A system's time response consists of two segments:

1. Transient response: The system's immediate reaction when the input changes, including overshoot, oscillation, and the speed of settling.
2. Steady-state response: The final constant value once the system stabilizes.

For any transfer function $G(s)$, the output is found by:

$$Y(s) = G(s)X(s)$$

where $X(s)$is the Laplace transform of the input.

1. Impulse Response

An impulse input has Laplace transform 1, so the output equals the transfer function itself. The impulse response reveals the inherent internal dynamics of the system without external shaping.

2. Step Response

A step input has Laplace transform 1/s. The step response is the most practical evaluation method in engineering because it exposes rise time, settling time, overshoot, damping, and general stability. These metrics are used in tuning control loops and reviewing system performance.

First-Order System Characteristics

A typical first-order element has the form:

$$G(s) = \frac{K}{1 + sT}$$

The step response:

$$y(t) = K(1 - e^{-t/T})$$

Key parameters:

- K: final gain
- T: time constant
- Settling time: approximately $4T$

This behavior is monotonic with no overshoot.

In power electronics, inductor current dynamics naturally follow first-order behavior, which becomes the plant model inside the current control loop.

Second-Order System Characteristics

The standard second-order transfer function is:

$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Parameters:

- ω_n: natural frequency
- ζ: damping ratio

System behavior depends on ζ:

- Overdamped ($\zeta > 1$): slow but stable, no overshoot
- Critically damped ($\zeta = 1$): fastest response without overshoot

- Underdamped ($0 < \zeta < 1$): overshoot and oscillation
- Undamped ($\zeta = 0$): continuous oscillation

These cases form the foundation for analyzing converter stability and controller tuning.

Buck Converter Dynamics (Plant Model)

For a current-controlled buck converter, the plant from duty cycle to inductor current is approximated as:

$$G_p(s) = \frac{1}{Ls + R}$$

If winding resistance is small:

$$G_p(s) \approx \frac{1}{Ls}$$

This is effectively a first-order or integrator-type system, consistent with classical first-order models.

The plant determines how quickly inductor current reacts to a change in duty cycle. The key physical parameter is:

- L: larger inductance slows current response
- R: contributes damping and limits speed

PI Controller and Closed-Loop Behavior

The PI controller used for current control is:

$$G_c(s) = K_p + \frac{K_i}{s}$$

Open-loop transfer function:

$$G_{ol}(s) = G_c(s)G_p(s)$$
$$= (K_p + \frac{K_i}{s})\frac{1}{Ls}$$

$$= \frac{K_p}{Ls} + \frac{K_i}{Ls^2}$$

This produces a second-order closed-loop system once feedback is applied. Matching coefficients with the standard second-order form yields:

$$\omega_n = \sqrt{\frac{K_i}{L}}, \zeta = \frac{K_p}{2}\sqrt{\frac{1}{LK_i}}$$

Meaning:

- $K_i$controls the loop speed and natural frequency
- $K_p$controls damping and overshoot
- Poor ratios between $K_p$and $K_i$create oscillation, sluggishness, or overshoot

This explains why the current loop in a buck converter behaves exactly like the textbook second-order system.

Step Response of the PI-Controlled Buck

A change in current reference produces a classical second-order step response:

- If $\zeta > 1$: slow, stable current rise
- If $\zeta = 1$: fastest clean tracking
- If $0 < \zeta < 1$: overshoot and ringing in the inductor current
- If $\zeta = 0$: persistent oscillation

These patterns match real converter waveforms. Incorrect gains lead to overshoot that stresses MOSFETs, increases losses, and destabilizes parallel operation.

Summary

The behavior of a current-controlled buck converter aligns directly with classical first-order and second-order system theory. The inductor defines a first-order plant, and the PI controller introduces second-order closed-loop dynamics. Impulse and step responses provide a clear framework for evaluating speed, damping, stability, and overshoot, making them essential tools for designing and tuning high-performance current control loops.